WEEK 3 PROJECT REPORT

A/B TESTING

Website Conversion Rate Analysis

Project Title	Website Conversion Rate Analysis Using A/B Testing
Tools Used	Python, Pandas, NumPy, Matplotlib, SciPy
Statistical Methods	Hypothesis Testing, Two-Proportion Z-Test, 95% Confidence Interval

1. INTRODUCTION

A/B Testing is a statistical testing method used to compare two different versions of a webpage, application, advertisement or product to identify which version performs better.

In A/B testing, users are divided into two groups:

- **Version A:** Existing version (Control Group).
- **Version B:** New or modified version (Treatment Group).

The performance of both versions is measured using a particular metric, such as conversion rate, click-through rate or sales.

In this project, we compare two website button designs to determine which one generates more purchases.

2. AIM

To analyse a sample A/B testing experiment using Python, compare two website versions, calculate conversion rates, perform hypothesis testing and state the final conclusion using a 95% confidence interval.

3. OBJECTIVES

1. Understand the concept of A/B testing.
2. Collect and analyse sample experimental data.
3. Calculate conversion rates for two groups.
4. Compare the performance of Version A and Version B.
5. Perform statistical hypothesis testing.
6. Calculate a 95% confidence interval.
7. Visualise the results using graphs.
8. Make a data-driven business decision.

4. TOOLS AND TECHNOLOGIES

Tool	Purpose
Python	Programming language
Pandas	Data manipulation
NumPy	Numerical calculations
Matplotlib	Data visualisation
SciPy	Statistical calculations
Google Colab	Running Python code

5. PROJECT SCENARIO

An online shopping company wants to improve its website purchase rate.

The company currently uses an existing Buy Now button (Version A). The company designs a new and more attractive Buy Now button (Version B).

The company conducts an experiment with 20,000 visitors to determine which button produces more purchases.

The visitors are divided randomly into two groups.

- Group A receives the existing button.
- Group B receives the new button.

The purchase conversion rates are compared to determine whether the new design improves performance.

6. DATASET DESCRIPTION

The following is an illustrative sample dataset created for this project.

Details	Version A	Version B
Total visitors	10,000	10,000
Purchases	1,000	1,120
Non-purchases	9,000	8,880
Conversion rate	10%	11.2%

Total visitors: 20,000

Total purchases: 2,120

7. UNDERSTANDING CONVERSION RATE

Conversion rate is the percentage of visitors who complete a desired action, such as purchasing a product.

> **Formula:**
> Conversion Rate = (Number of Conversions / Total Visitors) × 100

Version A: (1000 / 10000) × 100 = **10%**

Version B: (1120 / 10000) × 100 = **11.2%**

Therefore, Version B has a higher conversion rate.

8. HYPOTHESIS TESTING

Hypothesis testing is a statistical technique used to determine whether an observed difference between two groups is statistically significant.

Null Hypothesis (H0): There is no significant difference between the conversion rates of Version A and Version B.

Alternative Hypothesis (H1): There is a significant difference between the conversion rates of Version A and Version B.

Significance Level: Alpha = 0.05 (5%)

Decision rule:

- If p-value is less than 0.05, reject the null hypothesis.
- If p-value is greater than or equal to 0.05, fail to reject the null hypothesis.

9. STATISTICAL ANALYSIS

Absolute improvement: 11.2% - 10% = 1.2 percentage points.

Relative improvement: (11.2 - 10) / 10 × 100 = 12%.

Version B shows a 12% relative increase in conversion rate compared with Version A.

Two-Proportion Z-Test

The two-proportion Z-test is used to compare the conversion rates of two independent groups.

Calculated results:

- Z-statistic = 2.7564
- P-value = 0.005844

Since the p-value is less than 0.05, we reject the null hypothesis.

This indicates a statistically significant difference between the two conversion rates.

10. CONFIDENCE INTERVAL

A confidence interval is a range of values used to estimate the true difference between two population proportions.

Confidence Level = 95%

Difference (B - A) = 0.012.

95% Confidence Interval:
Lower limit = 0.00347
Upper limit = 0.02053

> Therefore: **95% Confidence Interval = +0.35 to +2.05 percentage points.**

Since the confidence interval does not contain zero, the result supports a positive improvement for Version B.

11. PYTHON IMPLEMENTATION

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from statsmodels.stats.proportion import proportions_ztest

# Step 1: Create sample dataset
data = pd.DataFrame({
    "Version": ["A", "B"],
    "Visitors": [10000, 10000],
    "Purchases": [1000, 1120]
})

# Step 2: Calculate conversion rate
data["Conversion_Rate"] = (
    data["Purchases"] / data["Visitors"]
) * 100

print("A/B Testing Dataset")
print(data)

# Step 3: Extract values
conversions = data["Purchases"].values
visitors = data["Visitors"].values

rate_A = conversions[0] / visitors[0]
rate_B = conversions[1] / visitors[1]

print("Version A:", rate_A * 100, "%")
print("Version B:", rate_B * 100, "%")

# Step 4: Calculate improvement
difference = rate_B - rate_A

print("Improvement:",
      difference * 100, "percentage points")

# Step 5: Perform hypothesis test
z_stat, p_value = proportions_ztest(
    conversions,
    visitors,
    alternative="two-sided"
)

print("Z Statistic:", z_stat)
print("P Value:", p_value)

# Step 6: Calculate 95% confidence interval
se = np.sqrt(
    rate_A * (1-rate_A) / visitors[0] +
    rate_B * (1-rate_B) / visitors[1]
)

lower = difference - 1.96 * se
upper = difference + 1.96 * se

print("95% Confidence Interval:")
print(lower * 100, "to", upper * 100)

# Step 7: Make conclusion
if p_value < 0.05:
    print("Reject Null Hypothesis")
    print("Version B shows a significant difference.")
else:
    print("Fail to Reject Null Hypothesis")

# Step 8: Visualize results
plt.bar(
    data["Version"],
    data["Conversion_Rate"]
)

plt.title("A/B Testing Conversion Rate")
plt.xlabel("Website Versions")
plt.ylabel("Conversion Rate (%)")
plt.show()
```

12. OUTPUT AND RESULTS

The experiment produces the following results:

Metric	Version A	Version B
Visitors	10,000	10,000
Purchases	1,000	1,120
Conversion rate	10%	11.2%

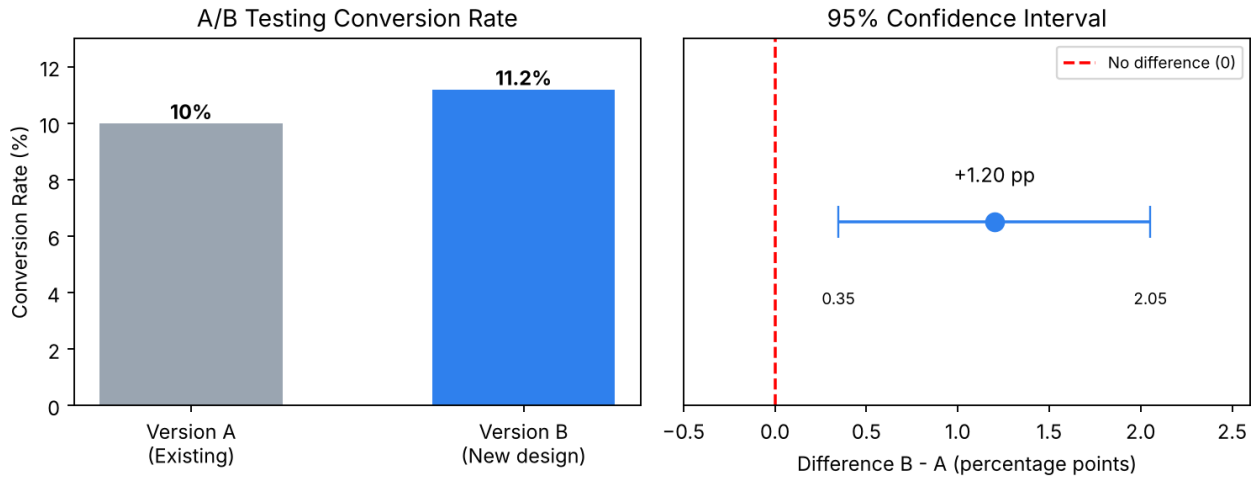
Statistical Results:

Metric	Result
Z-statistic	2.7564
P-value	0.005844
Confidence level	95%
Confidence interval	+0.35 to +2.05 percentage points
Decision	Reject H0

13. DATA VISUALIZATION

A bar chart (made with Matplotlib) compares the conversion rates of both website versions. The right-hand panel shows the 95% confidence interval for the difference.

- Version A conversion rate: 10%.
- Version B conversion rate: 11.2%.



The graph indicates that Version B performs better than Version A.

14. COMPARISON OF VERSION A AND VERSION B

Parameter	Version A	Version B
Button design	Existing	New design
Visitors	10,000	10,000
Purchases	1,000	1,120
Conversion rate	10%	11.2%
Performance	Lower	Higher

Version B performs better based on the sample results.

15. ADVANTAGES OF A/B TESTING

1. Helps make data-driven decisions.
2. Improves website performance.
3. Increases conversion rates.
4. Reduces guesswork.
5. Helps understand customer behaviour.
6. Supports better marketing strategies.
7. Helps identify effective designs and features.

16. APPLICATIONS OF A/B TESTING

- E-commerce websites.
- Digital marketing campaigns.
- Mobile applications.
- Email marketing.
- Social media advertisements.
- Website design optimisation.
- Online shopping platforms.
- User experience improvement.

17. LIMITATIONS

1. Requires sufficient sample size.
2. Experiments may take time.
3. Incorrect randomisation can create bias.
4. External factors can influence results.
5. Statistical significance does not always mean business success.
6. Poor experiment design can produce misleading conclusions.

18. FINAL CONCLUSION

In this project, A/B testing was performed to compare two different website button designs.

Version A achieved a conversion rate of 10%, whereas Version B achieved a conversion rate of 11.2%.

The experiment showed a 1.2 percentage-point improvement in conversion rate. The two-proportion Z-test produced a p-value of 0.005844, which is less than the significance level of 0.05.

The 95% confidence interval was approximately +0.35 to +2.05 percentage points.

> Therefore, we reject the null hypothesis and conclude that **Version B shows a statistically significant improvement over Version A** in this sample experiment.

This project helped demonstrate how Python and statistical methods can be used to analyse experiments and support data-driven business decisions.

19. WHAT I LEARNED

Through this project, I learned:

- The fundamentals of A/B testing.
- How to compare two versions of a product.
- How to calculate conversion rates.
- How to perform hypothesis testing.
- How to interpret p-values.
- How to calculate confidence intervals.
- How to visualise experimental results using Python.
- How to make conclusions based on statistical evidence.

20. VIVA QUESTIONS AND ANSWERS

1. What is A/B testing?

A/B testing is a method used to compare two versions of a product or webpage to identify which performs better.

2. What are Version A and Version B?

Version A is the existing version, and Version B is the modified version.

3. What is conversion rate?

It is the percentage of visitors who complete a desired action.

4. What is hypothesis testing?

It is a statistical method used to test assumptions using sample data.

5. What is a null hypothesis?

It states that there is no significant difference between two groups.

6. What is an alternative hypothesis?

It states that there is a significant difference between two groups.

7. What is a p-value?

It measures how extreme the observed result would be if the null hypothesis were true.

8. What is a confidence interval?

It is a range of values estimated using sample data to capture a population parameter with a stated long-run confidence level.

9. What is a Z-test?

It is a statistical test used to compare proportions or means under appropriate assumptions.

10. Which Python libraries are used?

Pandas, NumPy, Matplotlib and Statsmodels.

11. What is statistical significance?

It means the result meets a predefined statistical decision threshold.

12. What is the significance level in this project?

0.05 or 5%.

13. Which version performed better?

Version B, with an 11.2% conversion rate.

14. What is the confidence level used?

95%.

15. What is the final result?

Version B showed a statistically significant improvement in the sample experiment.

21. REFERENCES

1. Python Documentation – https://docs.python.org/
2. Pandas Documentation – https://pandas.pydata.org/docs/
3. NumPy Documentation – https://numpy.org/doc/
4. Matplotlib Documentation – https://matplotlib.org/
5. Statsmodels Documentation – https://www.statsmodels.org/

Note: *The dataset is illustrative sample data, not real company data.*